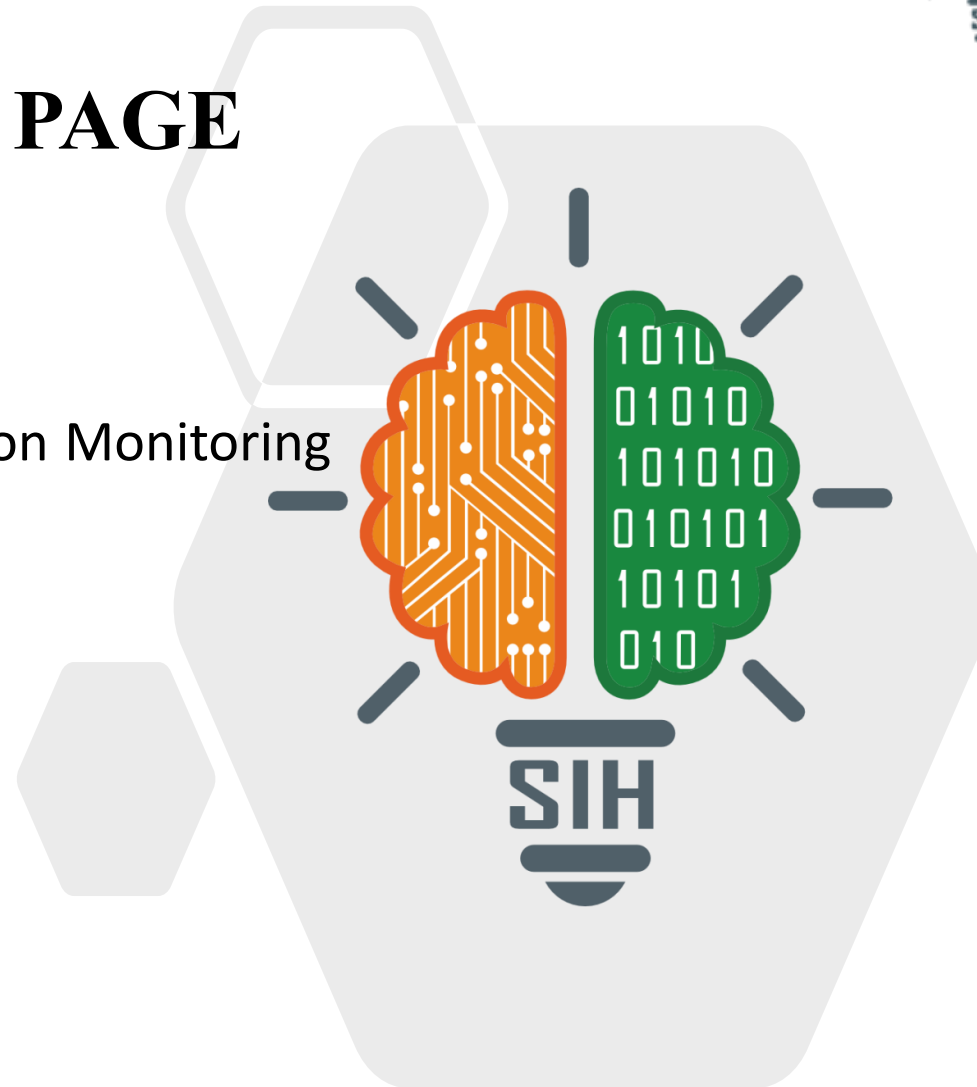


SMART INDIA HACKATHON 2025



TITLE PAGE

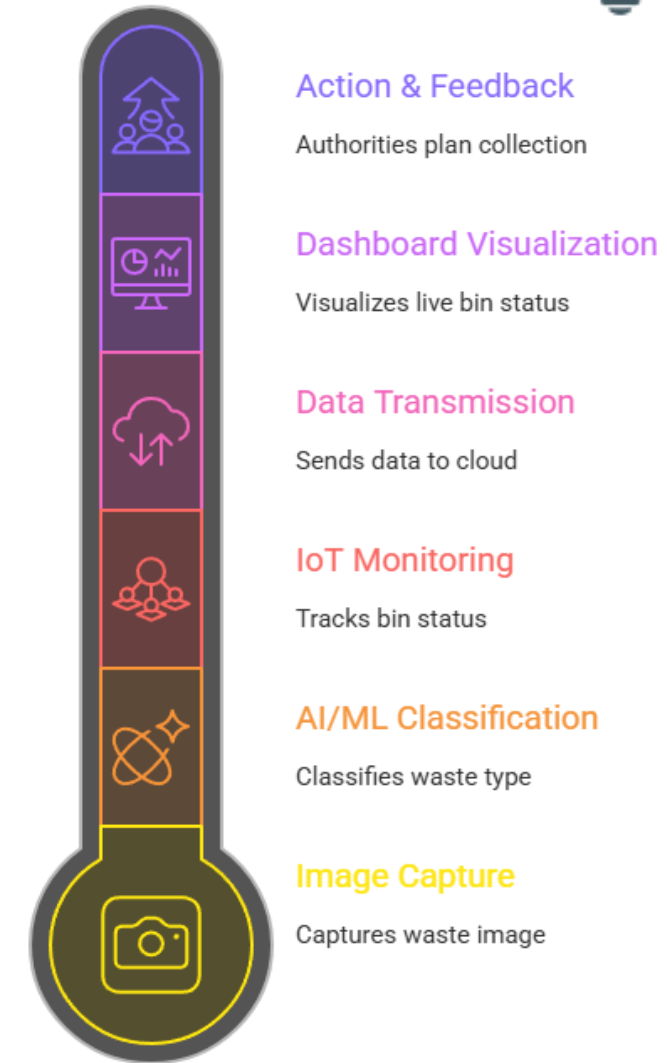
- **Problem Statement ID** – SIH25014
- **Problem Statement Title** - Waste Segregation Monitoring System for Urban Local Bodies.
- **Theme** - Clean & Green Technology.
- **PS Category**- Hardware
- **Team ID** -
- **Team Name (Registered on portal)**- Synapse



Proposed Solution :

- **Real time image capturing of waste materials.**
Camera captures the image of the waste materials in real time.
- **Waste Segregation using Machine Learning Model.**
AI model uses those real-time images and classifies biodegradable & non-biodegradable waste.
- **IoT-Enabled Smart Waste-bins.**
Sensors monitor fill level & bin status , enabling real-time updates.
- **Central Dashboard.**
For municipal corporation & waste collection managers shows live bin status, route optimization & compliance reports.

TECHNICAL APPROACH



Feasibility

- Low-cost solution.
- Rapid Prototyping.
- Provides Real-time Monitoring.
- Improved public sanitation.
- Scalable.

Challenges

- Classification error.
- Hardware failure.
- Public Apathy.
- Connectivity and Data Management.
- Lack of Standardized Policies.

Mitigation

- Enhancing AI accuracy.
- Robust Hardware Design.
- Spreading Public Awareness.
- Modular design.
- Scalable Cloud Infrastructure.

IMPACT AND BENEFITS



- Accurate waste segregation.
- Waste collection route optimization.
- Cost & time effective solution.
- Centralized dashboard for waste-monitoring.
- Improved public sanitation.

Future goals:

- Integration with smart city infrastructure.
- Enhanced Route Optimization.
- Forecasting waste generation trends.

RESEARCH AND REFERENCES



- Tensorflow Documentation.
- Kaggle Dataset.
- Raspberry Pi Documentation.
- React Bits.
- Node.js Documentation.
- Perplexity AI , Napkin AI.